



# Netflix Content Analysis Report

## 1. Project Overview

This project analyzes Netflix content data to uncover insights into content distribution, growth trends, ratings, and viewer targeting. The objective is to understand how Netflix has expanded its content library and identify key patterns that can support business and content strategy decisions.

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## 2. Dataset Summary

- **Source:** Netflix Titles Dataset
- **Rows:** ~8,800+ records
- **Columns:** Multiple attributes including content type, release year, rating, and duration

### Key Features:

- Content Type (Movie / TV Show)
- Title
- Release Year
- Date Added
- Rating
- Duration (Minutes / Seasons)
- Description

### Data Issues Identified:

- Missing values in rating, date\_added, and description
  - Inconsistent duration format
  - Mixed data types
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### 3. Data Cleaning & Preparation using Python

Data cleaning was performed using Python (Pandas).

#### ◆ Steps Performed:

#### ✓ Data Loading

- Loaded dataset using Pandas
  - Checked structure using `df.info()` and `df.describe()`
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#### ✓ Handling Missing Values

- Filled missing ratings with “**Not Rated**”
  - Replaced missing descriptions with **default text**
  - Imputed missing dates using **mode**
  - Removed rows with critical missing identifiers
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#### ✓ Data Transformation

- Converted `date_added` to datetime format
  - Extracted:
    - `year_added`
    - `month_added`
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#### ✓ Duration Handling

- Split duration into:
    - `duration_minutes` (Movies)
    - `duration_seasons` (TV Shows)
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#### ✓ Feature Engineering

- Created:
    - `content_age` → Age of content
    - `content_type_flag` → Movie (1), TV Show (0)
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## ✓ Data Type Corrections

- Converted:
    - `release_year` → Integer
    - `show_id` → String
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## ✓ Final Output

- Clean dataset exported as CSV for further analysis
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# 4. Data Analysis using SQL

The cleaned dataset was imported into SQL for structured analysis.

## 🔍 Key Business Questions Answered:

1. Distribution of Movies vs TV Shows
  2. Content growth over the years
  3. Most common content ratings
  4. Average movie duration
  5. Monthly content addition trends
  6. Top-performing release years
  7. Content age distribution
  8. Most recent content added
  9. Percentage share of Movies vs TV Shows
  10. Trend analysis using window functions
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## 💡 Example Insight:

- Movies dominate the platform (~68%)
  - Content growth accelerated after 2016
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## 5. Data Visualization using Tableau

An interactive dashboard was created in Tableau to present insights visually.

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### Dashboard Features:

- KPI Cards (Total Titles, Movies, TV Shows, Avg Duration)
  - Content Growth Trend (Line Chart)
  - Rating Distribution (Bar Chart)
  - Movies vs TV Shows (Pie Chart)
  - Global Distribution (Map)
  - Genre Analysis (Bar Chart)
  - Monthly Trend Analysis
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### Storytelling

A Tableau Story was created to present insights in a structured format:

- Overview
  - Growth Trend
  - Distribution
  - Content Characteristics
  - Key Business Insights
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## 6. Key Insights

- Movies account for the majority (~68%) of Netflix content
  - Content growth increased significantly after 2016
  - TV-MA and TV-14 are the most common ratings
  - Drama and International genres dominate
  - Most movies fall between 80–120 minutes
  - Netflix has a globally diverse content library
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## 7. Business Recommendations

- 🎬 Increase investment in high-performing genres (Drama, International)
  - 📈 Continue expanding content library based on growth trends
  - 🎯 Focus on mature audience targeting (TV-MA, TV-14)
  - 🌐 Expand global content production for wider reach
  - ⌚ Optimize movie duration for audience engagement
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## 8. Tools & Technologies Used

- **Python (Pandas, NumPy)** → Data Cleaning
  - **SQL (MySQL/PostgreSQL)** → Data Analysis
  - **Tableau** → Dashboard & Storytelling
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## 9. Conclusion

This project demonstrates an end-to-end data analysis workflow, including data cleaning, transformation, SQL-based analysis, and interactive visualization. The insights derived provide a clear understanding of Netflix's content strategy and growth patterns.

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